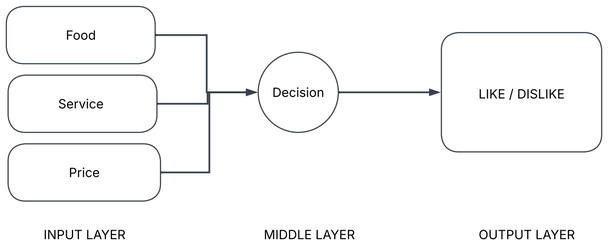
LAB 8 - How Neural Networks Learn
By: Danielle Lampitoc



ROLES - Individual Simulation

I will this activity individually by performing all three roles.

INPUT FEEDER

- Read each restaurant scenario.

DECISION NODE

- Predicts weather the customer will LIKE or DISLIKE the restaurant

OBSERVER / TRACKER

- Recorded predictions, checked correct outcomes, and adjusted featured weights after incorrect predictions.

Features	Weight (Stars)
Food	☆ ☆
Price	☆
Service	☆

PART A - TRAINING

Card	Prediction	Actual Answer	Result	Weight Change
1	LIKE	LIKE	Correct	None
2	DISLIKE	DISLIKE	Correct	None
3	LIKE	LIKE	Correct	None
4	DISLIKE	DISLIKE	Correct	None
5	LIKE	LIKE	Correct	None
6	DISLIKE	DISLIKE	Correct	None
7	LIKE	LIKE	Correct	None
8	DISLIKE	DISLIKE	Correct	None
9	DISLIKE	LIKE	Incorrect	Food +1 ☆
10	DISLIKE	DISLIKE	Correct	None

Training Results:

- During the training phase, I correctly predicted 9 of 10 restaurant scenarios, resulting in a 90% accuracy. My only incorrect prediciton, was on Card 9, where I had predicted DISLIKE rather than LIKE.
- Based on this result, I increased the weight for Food from 1 to 2 because it showed that food quality had a greater influence on customer satisfaction over service.

PART B - TESTING PHASE

Card	Prediction	Actual Answer	Result
11	LIKE	LIKE	Correct
12	DISLIKE	DISLIKE	Correct
13	LIKE	LIKE	Correct

Testing Results:

- During the testing phase, I correctly predicted all 3 restaurant scenarios achieving 100% accuracy. This was improvement from my training accuracy of 90%.
- By apply the adjusted weights, I was able to make accurate predictions on the scenarios. Food remained the most important feature, with a weight of 2, where Service and Price each had a weight of 1.

PART C - REFLECTION

How did your predictions improve?

During the training phase I had correctly predicted 9 out of 10 scenarios, achieving a 90% accuracy. After making an incorrect prediction on Card 9, I adjusted the weight of food to reflect its importance. During testing, I correctly predicted all three scenarios, increasing my accuracy to 100%.

How did weights (tokens) help you learn?

The weights helped me understand how much influence each feature had on the final decision. By using stars to represent weights, I was able to identify which features were most important. After my incorrect prediction, I increased Food's weight from 1 to 2 because food quality appeared to have a greater influence on customer satisfaction.

What would happen with more layers or inputs?

Adding more layers or inputs would allow neural networks to consider additional factors and recognize more complex patterns. For example, including cleanliness, atmosphere, or waiting times could help make more detailed predictions. However, this would also make the network more complex and require more training data.

How does this reflect how real AI works?

This simulation demonstrates how AI learns through predictions, feedback, and weight adjustments. Similar to this activity, real neural networks compare thier predictions with expected results and adjust their weights to reduce errors. Over time, this process helps AI recognize patterns and make more accurate decisions.